

Koushik GOPATHI

+91 89195 96047 | g.koushikvarma724@gmail.com | [LinkedIn](#) | github.com/Koushik-Gopathi | koushik-gopathi.vercel.app

PROFILE

Computer Science (Cyber Security) undergraduate at Amrita Vishwa Vidyapeetham who ships real software — full-stack PERN web apps, cross-platform Flutter apps, and AI-driven tools spanning voice assistants, on-device computer vision, and threat detection for LLM/MCP systems. ACM Student Chapter mentor focused on secure, user-centric products.

EDUCATION

B.Tech, CSE (Cyber Security) — Amrita Vishwa Vidyapeetham, Amritapuri CGPA: 7.61/10 | 2024 – Present
Intermediate (MPC) — Sri Chaitanya Junior College, Telangana 95.5% | 2022 – 2024
Secondary School (SSC, State Board) — Sri Chaitanya School, Telangana CGPA: 9.7/10 | 2021 – 2022

PROJECTS

AmbyoAI — Smartphone Amblyopia (Lazy Eye) Screening | *Flutter, ML Kit, TFLite, Vosk, FastAPI* Faculty-guided · Ongoing

- Offline, touchless, voice-driven eye-screening app for community health workers — runs fully on a smartphone; clinical validation with doctors is next.
- Implemented and calibrated the **Hirschberg corneal-reflex test** with ML Kit face-mesh; fixed Red Reflex, Gaze and Prism tests and built a light-mode UI.

Karen — Personal AI Voice Assistant | *Python/FastAPI, Flutter, Kotlin, Groq, OpenRouter (Claude), Deepgram* GitHub

- Built a cross-platform voice AI agent (Python/FastAPI + Flutter) with a 40-tool streaming agent loop, sub-second spoken responses, and barge-in interruption handling.
- Cut LLM cost ~10× with prompt caching and model routing that sends chat to free models and only real tasks to Claude.
- Shipped an Android app with its own agent runtime, foreground-service wake-word listening, and secure QR-paired LAN control of the desktop agent.

GuardianMesh — AI / MCP Threat Detection | *TypeScript, React, Vite* Hackathon Project

- Detects **16 threat categories** in AI/MCP interactions — prompt injection, command/SQL injection, XSS, credential & SSH-key leaks, phishing, supply-chain URLs.
- Scores intent, context, action verbs and permissions into a 0–100 risk score with confidence, recommendation and explanation.

LogBook — Attendance & Schedule Management | *React.js, Node.js, Express, PostgreSQL* ACM Student Chapter

- Co-developed a full-stack PERN app deployed for college use, with student-wise and course-wise attendance tracking, schedules and monthly reports.

Student Attendance Tracker | *Flutter, Dart, Firebase, GitHub Actions* Live Demo

- Cross-platform app for subject-wise attendance with per-student summaries; CI/CD auto-deploys to GitHub Pages.

Other Projects: Personal Portfolio (Next.js, TypeScript, Framer Motion) • NASA APOD Explorer (React.js, NASA API) • Kargil Diwas Marathon Registration Site

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, TypeScript, Dart, SQL, HTML, CSS, Shell
Web & Mobile: React.js, Next.js, Node.js, Express, REST APIs, FastAPI, Flutter, Framer Motion
AI / ML: LLM APIs & tool calling, Speech-to-Text / Text-to-Speech, Google ML Kit, TensorFlow Lite
Databases & Tools: PostgreSQL, Firebase, Git, GitHub, GitHub Actions, VS Code, Linux
Security: Web vulnerabilities (SQLi, XSS, command injection), prompt-injection detection, threat analysis

LEADERSHIP & COMMUNITY

- **Mentor, ACM Student Chapter, Amrita Amritapuri (2026)** — conducted the junior induction session (Jul 2026); guide juniors in web & app development.
- **Kargil Vijay Diwas Marathon (Jul 2026)** — co-led organisation for **400+ participants**, handling end-to-end logistics.
- **Vishmaya Summer Camp (May 2026)** — taught hands-on web development basics to school students.
- **Volunteer** — Amrita Kalotsavam, Arts & Crafts (Feb 2026); Campus Cleanup Drive, Amritapuri (2025).

CERTIFICATIONS

- Introduction to Cybersecurity — Cisco
- Oracle Database SQL (DBMS) — Oracle
- Professional Soft Skills Pathway — LinkedIn Learning
- Critical Thinking & Decision-Making — LinkedIn Learning

LANGUAGES & HOBBIES

- English (Professional), Telugu (Native)
- Hindi (Basic), German (Beginner)
- **Hobbies:** Sketching, Gaming